The Galois Field Mechanic Transform:

Discrete Phase Spaces, Quantum Kinematics, and Finite Field Cosmologies

A Rigorous Mathematical Formulation with Complete Analytical Proofs

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Abstract

Continuous-variable formulations of quantum mechanics built upon infinite-dimensional complex Hilbert spaces $\mathcal{H} = L^2(\mathbb{R})$ suffer from topological and operational mismatches when applied to strictly finite-dimensional quantum processors, discrete quantum state tomography, and finite-entropy cosmologies. In this treatise, we establish the complete, rigorous mathematical foundation of the *Galois Field Mechanic Transform* (GFMT). We develop finite field arithmetic over Galois extensions $\mathbb{F}_q = \mathrm{GF}(p^n)$, the absolute field trace, and additive character harmonic analysis. We define the Galois Field Fourier Transform (GFFT) as a unitary operator acting on the finite Hilbert space $\mathcal{H}_q \cong \mathbb{C}^q$, proving its unitarity, spectral order, and Plancherel isometry. We construct the discrete Weyl-Heisenberg operator algebra, prove the canonical Schwinger-Weyl commutation relations, and map kinematic observables onto a $q \times q$ discrete affine phase space. Furthermore, we construct complete sets of $q+1$ Mutually Unbiased Bases (MUBs) directly from affine striations, formalize Stratonovich-Weyl phase space point operators, and prove the marginality and tracial orthogonality of the Discrete Wigner Function. We bridge this discrete framework to fault-tolerant quantum computing via the $\mathbb{F}_4$-symplectic isomorphism for multi-qubit Pauli stabilizers, analyze hardware-accelerated Reed-Solomon coding, and detail non-Archimedean p-adic transforms alongside Felix Lev's Galois Field Quantum Theory (GFQT) de Sitter cosmologies.

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1 Introduction and Algebraic Foundations of Galois Fields

Standard continuous quantum mechanics represents canonical coordinates (position x and momentum p) on the continuous real phase space $\mathbb{R}^{2n}$, generating infinite-dimensional Hilbert spaces. For discrete quantum registers—such as qudits of dimension $d = p^n$ (where p is a prime characteristic and $n \in \mathbb{N}^+$)—the continuous manifold is geometrically inapplicable. A rigorous finite kinematic theory requires that coordinate configurations belong to a finite algebraic field where division by non-zero elements is globally defined.

Definition 1.1 (Galois Field Extension)**.** Let p be a prime number and n a positive integer. The Galois field $\mathbb{F}_q$ of order $q = p^n$ is the unique finite field containing q elements. Its prime subfield is isomorphic to the integers modulo p:

$$\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}. \tag{1}$$

The extension field $\mathbb{F}_q$ is algebraically constructed as the quotient ring:

$$\mathbb{F}_q \cong \mathbb{F}_p[x]/\langle P(x)\rangle, \tag{2}$$

where $P(x) \in \mathbb{F}_p[x]$ is a monic irreducible polynomial of degree $\deg(P) = n$.

Let $\theta \in \mathbb{F}_q$ be a designated root of $P(x)$, so that $P(\theta) = 0$. The set $\{1, \theta, \theta^2, \ldots, \theta^{n-1}\}$ forms a polynomial basis for $\mathbb{F}_q$ viewed as an n-dimensional vector space over $\mathbb{F}_p$. Any arbitrary field element $\alpha \in \mathbb{F}_q$ has a unique expansion:

$$\alpha = \sum_{j=0}^{n-1} a_j \theta^j, \quad a_j \in \mathbb{F}_p. \tag{3}$$

Field addition is defined componentwise modulo p, and field multiplication is executed by polynomial multiplication modulo $P(\theta)$.

Definition 1.2 (Frobenius Automorphism)**.** The Frobenius map $\sigma : \mathbb{F}_q \to \mathbb{F}_q$ is defined by:

$$\sigma(\alpha) = \alpha^p. \tag{4}$$

The Galois group $\mathrm{Gal}(\mathbb{F}_q/\mathbb{F}_p)$ is a cyclic group of order n generated by σ:

$$\mathrm{Gal}(\mathbb{F}_q/\mathbb{F}_p) = \langle\sigma\rangle = \{\mathrm{id}, \sigma, \sigma^2, \ldots, \sigma^{n-1}\}. \tag{5}$$

The fixed field of σ is precisely the prime field: $\mathbb{F}_q^\sigma = \{\alpha \in \mathbb{F}_q : \alpha^p = \alpha\} = \mathbb{F}_p$.

2 The Absolute Galois Trace and Additive Harmonic Analysis

To construct Fourier analysis on $\mathbb{F}_q$, one requires a bridge between field elements in $\mathbb{F}_q$ and the unit circle in $\mathbb{C}$. This is achieved via the absolute trace map.

Definition 2.1 (Absolute Galois Trace)**.** For any $\alpha \in \mathbb{F}_q$, the absolute trace $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is defined as the sum of all Galois conjugates of α:

$$\mathrm{Tr}(\alpha) \equiv \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(\alpha) = \sum_{k=0}^{n-1} \sigma^k(\alpha) = \sum_{k=0}^{n-1} \alpha^{p^k} = \alpha + \alpha^p + \alpha^{p^2} + \cdots + \alpha^{p^{n-1}}. \tag{6}$$

Lemma 2.1 (Prime Subfield Confinement)**.** *For every $\alpha \in \mathbb{F}_q$, $\mathrm{Tr}(\alpha) \in \mathbb{F}_p$.*

Proof. An element $\xi \in \mathbb{F}_q$ lies in the prime field $\mathbb{F}_p$ if and only if $\sigma(\xi) = \xi^p = \xi$. Applying the Frobenius automorphism to $\mathrm{Tr}(\alpha)$ and utilizing the algebraic freshman's dream identity $(x+y)^p = x^p + y^p$ in characteristic p:

$$\begin{aligned}(\mathrm{Tr}(\alpha))^p &= \left(\sum_{k=0}^{n-1} \alpha^{p^k}\right)^p = \sum_{k=0}^{n-1} (\alpha^{p^k})^p = \sum_{k=0}^{n-1} \alpha^{p^{k+1}} \\ &= \alpha^p + \alpha^{p^2} + \cdots + \alpha^{p^{n-1}} + \alpha^{p^n}. \end{aligned} \tag{7}$$

Because every element $\alpha \in \mathbb{F}_q$ satisfies the field identity $\alpha^{p^n} = \alpha^q = \alpha$, the final term wraps around:

$$(\mathrm{Tr}(\alpha))^p = \alpha + \alpha^p + \cdots + \alpha^{p^{n-1}} = \mathrm{Tr}(\alpha). \tag{8}$$

Hence $\mathrm{Tr}(\alpha) \in \mathbb{F}_p$. □

Lemma 2.2 ($\mathbb{F}_p$-Linearity of the Trace)**.** *The absolute trace is an $\mathbb{F}_p$-linear functional: for all $c_1, c_2 \in \mathbb{F}_p$ and $\alpha, \beta \in \mathbb{F}_q$,*

$$\mathrm{Tr}(c_1\alpha + c_2\beta) = c_1\mathrm{Tr}(\alpha) + c_2\mathrm{Tr}(\beta). \tag{9}$$

Proof. Since $c_i \in \mathbb{F}_p$, $c_i^{p^k} = c_i$ for all $k \geq 0$. By the additivity of the Frobenius map:

$$\begin{aligned}\mathrm{Tr}(c_1\alpha + c_2\beta) &= \sum_{k=0}^{n-1} (c_1\alpha + c_2\beta)^{p^k} = \sum_{k=0}^{n-1} \left(c_1^{p^k}\alpha^{p^k} + c_2^{p^k}\beta^{p^k}\right) \\ &= c_1 \sum_{k=0}^{n-1} \alpha^{p^k} + c_2 \sum_{k=0}^{n-1} \beta^{p^k} = c_1\mathrm{Tr}(\alpha) + c_2\mathrm{Tr}(\beta). \end{aligned} \tag{10}$$

□

Lemma 2.3 (Non-Degeneracy of the Trace Form)**.** *The symmetric bilinear pairing $B : \mathbb{F}_q \times \mathbb{F}_q \to \mathbb{F}_p$ defined by $B(\alpha, \beta) = \mathrm{Tr}(\alpha\beta)$ is non-degenerate. That is, if $\mathrm{Tr}(\alpha\beta) = 0$ for all $\beta \in \mathbb{F}_q$, then $\alpha = 0$.*

Proof. Consider the polynomial $T(x) = \sum_{k=0}^{n-1} x^{p^k} \in \mathbb{F}_p[x]$. Its degree is $\deg(T) = p^{n-1} < p^n = q$. By Lagrange's theorem on polynomial roots, $T(x)$ can possess at most p^{n-1} roots in $\mathbb{F}_q$. Since $q = p^n > p^{n-1}$, there exists at least one element $\xi_0 \in \mathbb{F}_q$ such that $T(\xi_0) = \mathrm{Tr}(\xi_0) \neq 0$.

Now suppose $\alpha \in \mathbb{F}_q \setminus \{0\}$. Since $\mathbb{F}_q$ is a field, the multiplication map $m_\alpha : \beta \mapsto \alpha\beta$ is an automorphism of the additive group of $\mathbb{F}_q$. Its image is all of $\mathbb{F}_q$. Therefore, choosing $\beta = \alpha^{-1}\xi_0 \in \mathbb{F}_q$ yields:

$$\mathrm{Tr}(\alpha\beta) = \mathrm{Tr}\left(\alpha(\alpha^{-1}\xi_0)\right) = \mathrm{Tr}(\xi_0) \neq 0. \tag{11}$$

Thus, the radical of the trace bilinear form is trivial: $\ker(B) = \{0\}$. □

Definition 2.2 (Additive Character)**.** Let $\omega_p = \exp(2\pi i/p)$ denote the primitive complex p-th root of unity. For any $u \in \mathbb{F}_q$, the additive character $\chi_u : \mathbb{F}_q \to \mathbb{C}^\times$ is defined by:

$$\chi_u(\alpha) \equiv \chi(u\alpha) = \omega_p^{\mathrm{Tr}(u\alpha)} = \exp\left(\frac{2\pi i}{p}\mathrm{Tr}(u\alpha)\right). \tag{12}$$

Theorem 2.4 (Character Homomorphism and Dual Group Isomorphism)**.** *For each $u \in \mathbb{F}_q$, χ_u is a group homomorphism from the additive group $(\mathbb{F}_q, +)$ to the circle group $(S^1, \cdot)$. Furthermore, the map $\Phi : u \mapsto \chi_u$ is an isomorphism between $(\mathbb{F}_q, +)$ and its Pontryagin dual group $\widehat{\mathbb{F}_q}$.*

Proof. By the linearity established in Lemma 2.2:

$$\begin{aligned}\chi_u(\alpha+\beta) &= \omega_p^{\mathrm{Tr}(u(\alpha+\beta))} = \omega_p^{\mathrm{Tr}(u\alpha)+\mathrm{Tr}(u\beta)} \\ &= \omega_p^{\mathrm{Tr}(u\alpha)}\omega_p^{\mathrm{Tr}(u\beta)} = \chi_u(\alpha)\chi_u(\beta).\end{aligned} \tag{13}$$

Hence χ_u is a homomorphism. To show $\Phi : u \mapsto \chi_u$ is injective, examine its kernel:

$$\ker(\Phi) = \{u \in \mathbb{F}_q : \chi_u(\alpha) = 1 \quad \forall \alpha \in \mathbb{F}_q\}. \tag{14}$$

$\chi_u(\alpha) = 1$ implies $\mathrm{Tr}(u\alpha) \equiv 0 \pmod p$ for all $\alpha \in \mathbb{F}_q$. Since $\mathrm{Tr}(u\alpha) \in \mathbb{F}_p$, this means $\mathrm{Tr}(u\alpha) = 0$ for all $\alpha \in \mathbb{F}_q$. By Lemma 2.3, this forces $u = 0$. Thus $\ker(\Phi) = \{0\}$, establishing injectivity. Since $|\mathbb{F}_q| = q = |\widehat{\mathbb{F}_q}|$, the pigeonhole principle guarantees that Φ is an isomorphism. □

Theorem 2.5 (Fundamental Orthogonality Identity)**.** *For any $u \in \mathbb{F}_q$:*

$$\sum_{\alpha\in\mathbb{F}_q} \chi(u\alpha) = q\,\delta_{u,0}. \tag{15}$$

Proof. If $u = 0$, then for all $\alpha \in \mathbb{F}_q$, $\chi(0 \cdot \alpha) = \chi(0) = \omega_p^{\mathrm{Tr}(0)} = \omega_p^0 = 1$. The sum reduces to:

$$\sum_{\alpha\in\mathbb{F}_q} 1 = q. \tag{16}$$

If $u \neq 0$, by Theorem 2.4 the character χ_u is non-trivial, meaning there exists $\beta_0 \in \mathbb{F}_q$ such that $\chi(u\beta_0) \neq 1$. Multiplying the sum $S = \sum_{\alpha\in\mathbb{F}_q} \chi(u\alpha)$ by $\chi(u\beta_0)$:

$$\chi(u\beta_0)S = \chi(u\beta_0)\sum_{\alpha\in\mathbb{F}_q}\chi(u\alpha) = \sum_{\alpha\in\mathbb{F}_q}\chi(u(\alpha+\beta_0)). \tag{17}$$

As α ranges over all elements of $\mathbb{F}_q$, the shifted element $\alpha' = \alpha + \beta_0$ also traverses $\mathbb{F}_q$ bijectively. Hence:

$$\chi(u\beta_0)S = \sum_{\alpha'\in\mathbb{F}_q}\chi(u\alpha') = S \implies (1-\chi(u\beta_0))\,S = 0. \tag{18}$$

Because $\chi(u\beta_0) \neq 1$, we have $1 - \chi(u\beta_0) \neq 0$, which strictly forces $S = 0$. □

3 The Galois Field Fourier Transform (GFFT) Operator

Equipped with character orthogonality, we construct the kinematic transform between discrete position and momentum bases in quantum mechanics.

Definition 3.1 (State Space and Computational Basis)**.** Let $\mathcal{H}_q \cong \mathbb{C}^q$ denote a q-dimensional complex Hilbert space. We assign an orthonormal computational basis labeled by the field elements $\alpha \in \mathbb{F}_q$:

$$\mathcal{B}_{\text{pos}} = \{|\alpha\rangle : \alpha \in \mathbb{F}_q\}, \quad \langle\alpha|\beta\rangle = \delta_{\alpha,\beta}, \quad \sum_{\alpha\in\mathbb{F}_q} |\alpha\rangle\langle\alpha| = I_q. \tag{19}$$

We designate $\mathcal{B}_{\text{pos}}$ as the discrete *position basis*.

Definition 3.2 (Galois Field Fourier Transform)**.** The Galois Field Fourier Transform (GFFT) is the linear operator $\mathcal{F}_q : \mathcal{H}_q \to \mathcal{H}_q$ defined by:

$$\mathcal{F}_q \equiv \frac{1}{\sqrt{q}} \sum_{\alpha,\beta\in\mathbb{F}_q} \chi(\alpha\beta)|\beta\rangle\langle\alpha|. \tag{20}$$

The discrete *momentum basis* $\{|\tilde{\beta}\rangle : \beta \in \mathbb{F}_q\}$ is defined as the image of the position basis under $\mathcal{F}_q$:

$$|\tilde{\beta}\rangle \equiv \mathcal{F}_q|\beta\rangle = \frac{1}{\sqrt{q}} \sum_{\alpha\in\mathbb{F}_q} \chi(\alpha\beta)|\alpha\rangle. \tag{21}$$

Theorem 3.1 (Unitarity of the GFFT Operator)**.** *The operator $\mathcal{F}_q$ is unitary:*

$$\mathcal{F}_q^\dagger \mathcal{F}_q = \mathcal{F}_q \mathcal{F}_q^\dagger = I_q. \tag{22}$$

Proof. Compute the adjoint $\mathcal{F}_q^\dagger$:

$$\begin{aligned}\mathcal{F}_q^\dagger &= \left(\frac{1}{\sqrt{q}} \sum_{\mu,\nu\in\mathbb{F}_q} \chi(\mu\nu)|\nu\rangle\langle\mu|\right)^\dagger = \frac{1}{\sqrt{q}} \sum_{\mu,\nu\in\mathbb{F}_q} \chi(\mu\nu)^*|\mu\rangle\langle\nu| \\ &= \frac{1}{\sqrt{q}} \sum_{\mu,\nu\in\mathbb{F}_q} \chi(-\mu\nu)|\mu\rangle\langle\nu|.\end{aligned} \tag{23}$$

Now evaluate the product $\mathcal{F}_q\mathcal{F}_q^\dagger$:

$$\begin{aligned}\mathcal{F}_q\mathcal{F}_q^\dagger &= \left(\frac{1}{\sqrt{q}} \sum_{\alpha,\beta\in\mathbb{F}_q} \chi(\alpha\beta)|\beta\rangle\langle\alpha|\right)\left(\frac{1}{\sqrt{q}} \sum_{\mu,\nu\in\mathbb{F}_q} \chi(-\mu\nu)|\mu\rangle\langle\nu|\right) \\ &= \frac{1}{q} \sum_{\alpha,\beta,\mu,\nu\in\mathbb{F}_q} \chi(\alpha\beta)\chi(-\mu\nu)|\beta\rangle \underbrace{\langle\alpha|\mu\rangle}_{\delta_{\alpha,\mu}}\langle\nu| \\ &= \frac{1}{q} \sum_{\beta,\nu\in\mathbb{F}_q} |\beta\rangle\langle\nu| \sum_{\alpha\in\mathbb{F}_q} \chi(\alpha\beta)\chi(-\alpha\nu) \\ &= \frac{1}{q} \sum_{\beta,\nu\in\mathbb{F}_q} |\beta\rangle\langle\nu| \left(\sum_{\alpha\in\mathbb{F}_q} \chi\left(\alpha(\beta-\nu)\right)\right).\end{aligned} \tag{24}$$

Applying the Fundamental Orthogonality Theorem (Theorem 2.5):

$$\sum_{\alpha\in\mathbb{F}_q} \chi\left(\alpha(\beta-\nu)\right) = q\,\delta_{\beta-\nu,0} = q\,\delta_{\beta,\nu}. \tag{25}$$

Substituting this back into the expansion:

$$\mathcal{F}_q\mathcal{F}_q^\dagger = \frac{1}{q} \sum_{\beta,\nu\in\mathbb{F}_q} |\beta\rangle\langle\nu|\,(q\,\delta_{\beta,\nu}) = \sum_{\beta\in\mathbb{F}_q} |\beta\rangle\langle\beta| = I_q. \tag{26}$$

An identical derivation evaluates $\mathcal{F}_q^\dagger\mathcal{F}_q = I_q$. Hence $\mathcal{F}_q$ is strictly unitary. □

Theorem 3.2 (Mutual Unbiasedness of Position and Momentum)**.** *The position basis $\mathcal{B}_{\text{pos}} = \{|\alpha\rangle\}$ and the momentum basis $\mathcal{B}_{\text{mom}} = \{|\tilde{\beta}\rangle\}$ are mutually unbiased:*

$$|\langle\alpha|\tilde{\beta}\rangle|^2 = \frac{1}{q}, \quad \forall\alpha,\beta \in \mathbb{F}_q. \tag{27}$$

Proof. Compute the inner product:

$$\langle\alpha|\tilde{\beta}\rangle = \left\langle \alpha \middle| \left(\frac{1}{\sqrt{q}} \sum_{\xi\in\mathbb{F}_q} \chi(\xi\beta)|\xi\rangle \right) \right\rangle = \frac{1}{\sqrt{q}} \sum_{\xi\in\mathbb{F}_q} \chi(\xi\beta) \underbrace{\langle\alpha|\xi\rangle}_{\delta_{\alpha,\xi}} = \frac{1}{\sqrt{q}}\chi(\alpha\beta). \tag{28}$$

Taking the squared modulus:

$$|\langle\alpha|\tilde{\beta}\rangle|^2 = \frac{1}{q}|\chi(\alpha\beta)|^2 = \frac{1}{q}\left|\exp\left(\frac{2\pi i}{p}\mathrm{Tr}(\alpha\beta)\right)\right|^2 = \frac{1}{q}\cdot 1 = \frac{1}{q}. \tag{29}$$

□

Theorem 3.3 (Spectral Order and Parity Operator)**.** *The square of the GFFT operator is the discrete parity (coordinate reflection) operator $\mathcal{P} \equiv \mathcal{F}_q^2$:*

$$\mathcal{F}_q^2|\alpha\rangle = |-\alpha\rangle, \quad \forall\alpha \in \mathbb{F}_q. \tag{30}$$

Consequently, $\mathcal{F}_q^4 = I_q$, so the eigenvalues of $\mathcal{F}_q$ are restricted to the fourth roots of unity $\{\pm 1, \pm i\}$. Moreover, if $p = 2$, $-1 = 1$ in $\mathbb{F}_{2^n}$, which implies $\mathcal{F}_q^2 = I_{2^n}$ and $\mathcal{F}_q = \mathcal{F}_q^\dagger$ is Hermitian.

Proof. Evaluate $\mathcal{F}_q^2$ on a position state $|\alpha\rangle$:

$$\begin{aligned}\mathcal{F}_q^2|\alpha\rangle &= \mathcal{F}_q\left(\frac{1}{\sqrt{q}}\sum_{\beta\in\mathbb{F}_q}\chi(\alpha\beta)|\beta\rangle\right) = \frac{1}{q}\sum_{\beta\in\mathbb{F}_q}\chi(\alpha\beta)\left(\sum_{\gamma\in\mathbb{F}_q}\chi(\beta\gamma)|\gamma\rangle\right) \\ &= \frac{1}{q}\sum_{\gamma\in\mathbb{F}_q}|\gamma\rangle\left(\sum_{\beta\in\mathbb{F}_q}\chi(\beta(\alpha+\gamma))\right).\end{aligned} \tag{31}$$

By Theorem 2.5, the inner sum evaluates to $q\,\delta_{\alpha+\gamma,0} = q\,\delta_{\gamma,-\alpha}$. Therefore:

$$\mathcal{F}_q^2|\alpha\rangle = \frac{1}{q}\sum_{\gamma\in\mathbb{F}_q}|\gamma\rangle\,(q\,\delta_{\gamma,-\alpha}) = |-\alpha\rangle. \tag{32}$$

Applying $\mathcal{F}_q^2$ twice yields:

$$\mathcal{F}_q^4|\alpha\rangle = \mathcal{F}_q^2|-\alpha\rangle = |-(-\alpha)\rangle = |\alpha\rangle \implies \mathcal{F}_q^4 = I_q. \tag{33}$$

In characteristic $p = 2$, $-\alpha = \alpha$ for all $\alpha \in \mathbb{F}_{2^n}$, so $\mathcal{F}_q^2 = I_q$, proving that $\mathcal{F}_q$ is an involution and Hermitian. □

4 Discrete Weyl-Heisenberg Group and Operator Kinematics

In continuum quantum mechanics, position shifts are generated by momentum and momentum boosts are generated by position. In the Galois framework, these translations are realized by discrete unitary shift and clock operators.

Definition 4.1 (Kinematical Shift and Clock Operators)**.** For any field displacement parameters $\alpha, \beta \in \mathbb{F}_q$, define:

$$X_\alpha \equiv \sum_{\xi \in \mathbb{F}_q} |\xi + \alpha\rangle\langle\xi| \quad \text{(Position Shift Operator)}, \tag{34}$$

$$Z_\beta \equiv \sum_{\xi \in \mathbb{F}_q} \chi(\beta\xi)|\xi\rangle\langle\xi| \quad \text{(Momentum Shift / Clock Operator)}. \tag{35}$$

Theorem 4.1 (Schwinger-Weyl Canonical Commutation Relations)**.** *For all $\alpha, \beta \in \mathbb{F}_q$, the operators X_α and Z_β satisfy:*

$$Z_\beta X_\alpha = \chi(\alpha\beta) X_\alpha Z_\beta. \tag{36}$$

Proof. Let $|\xi\rangle$ be an arbitrary position basis state. Compute the action of the left-hand side:

$$Z_\beta X_\alpha |\xi\rangle = Z_\beta |\xi + \alpha\rangle = \chi(\beta(\xi + \alpha))|\xi + \alpha\rangle = \chi(\alpha\beta)\chi(\beta\xi)|\xi + \alpha\rangle. \tag{37}$$

Now compute the action of the right-hand side operator on $|\xi\rangle$:

$$X_\alpha Z_\beta |\xi\rangle = X_\alpha \left(\chi(\beta\xi)|\xi\rangle\right) = \chi(\beta\xi) X_\alpha |\xi\rangle = \chi(\beta\xi)|\xi + \alpha\rangle. \tag{38}$$

Multiplying the right-hand result by $\chi(\alpha\beta)$ exactly matches the left-hand side for every basis state $|\xi\rangle$. Since this holds over a complete basis, the operator identity $Z_\beta X_\alpha = \chi(\alpha\beta) X_\alpha Z_\beta$ holds identically. □

Theorem 4.2 (Fourier Duality of Kinematical Shifts)**.** *The GFFT operator $\mathcal{F}_q$ acts as the canonical intertwiner between position shifts and clock phase shifts:*

$$\mathcal{F}_q X_\alpha \mathcal{F}_q^\dagger = Z_{-\alpha} = Z_\alpha^\dagger, \quad \mathcal{F}_q Z_\beta \mathcal{F}_q^\dagger = X_\beta. \tag{39}$$

Proof. We evaluate the first conjugation on an arbitrary state $|\xi\rangle$:

$$\begin{aligned} \mathcal{F}_q X_\alpha \mathcal{F}_q^\dagger |\xi\rangle &= \mathcal{F}_q X_\alpha \left(\frac{1}{\sqrt{q}} \sum_{\gamma \in \mathbb{F}_q} \chi(-\xi\gamma)|\gamma\rangle \right) = \frac{1}{\sqrt{q}} \sum_{\gamma \in \mathbb{F}_q} \chi(-\xi\gamma) \mathcal{F}_q |\gamma + \alpha\rangle \\ &= \frac{1}{q} \sum_{\gamma, \mu \in \mathbb{F}_q} \chi(-\xi\gamma)\chi((\gamma + \alpha)\mu)|\mu\rangle \\ &= \frac{1}{q} \sum_{\mu \in \mathbb{F}_q} \chi(\alpha\mu)|\mu\rangle \left(\sum_{\gamma \in \mathbb{F}_q} \chi(\gamma(\mu - \xi)) \right). \end{aligned} \tag{40}$$

By Theorem 2.5, the inner sum over γ is $q\,\delta_{\mu,\xi}$. Thus:

$$\mathcal{F}_q X_\alpha \mathcal{F}_q^\dagger |\xi\rangle = \chi(\alpha\xi)|\xi\rangle = \chi(-(-\alpha)\xi)|\xi\rangle = Z_{-\alpha}|\xi\rangle. \tag{41}$$

For the second relation, recall from Theorem 3.3 that $\mathcal{F}_q^2 = \mathcal{P}$, where $\mathcal{P}|\xi\rangle = |-\xi\rangle$. Conjugating X_β by $\mathcal{F}_q^2$ gives $\mathcal{P} X_\beta \mathcal{P} = X_{-\beta}$. Substituting $Z_{-\alpha} = \mathcal{F}_q X_\alpha \mathcal{F}_q^\dagger$:

$$\mathcal{F}_q Z_\beta \mathcal{F}_q^\dagger = \mathcal{F}_q (\mathcal{F}_q X_{-\beta} \mathcal{F}_q^\dagger) \mathcal{F}_q^\dagger = \mathcal{F}_q^2 X_{-\beta} (\mathcal{F}_q^2)^\dagger = \mathcal{P} X_{-\beta} \mathcal{P} = X_\beta. \tag{42}$$

□

Definition 4.2 (Generalized Phase Space Displacement Operators)**.** For odd prime characteristic $p > 2$, $2 \in \mathbb{F}_p^\times$ possesses a unique multiplicative inverse $2^{-1} = \frac{p+1}{2} \pmod p$. For any phase space coordinate $(\alpha, \beta) \in \mathbb{F}_q \times \mathbb{F}_q$, the Weyl displacement operator is defined by:

$$D(\alpha, \beta) \equiv \chi\left(-2^{-1}\alpha\beta\right) X_\alpha Z_\beta. \tag{43}$$

Theorem 4.3 (Displacement Group Composition Law)**.** *For odd p, the displacement operators obey the projective group composition law:*

$$D(\alpha_1,\beta_1)D(\alpha_2,\beta_2) = \chi\left(2^{-1}(\alpha_1\beta_2 - \alpha_2\beta_1)\right) D(\alpha_1+\alpha_2, \beta_1+\beta_2). \tag{44}$$

Proof. Using Definition 4.2 and Theorem 4.1:

$$\begin{aligned} D(\alpha_1,\beta_1)D(\alpha_2,\beta_2) &= \chi\left(-2^{-1}\alpha_1\beta_1\right)\chi\left(-2^{-1}\alpha_2\beta_2\right) X_{\alpha_1}Z_{\beta_1}X_{\alpha_2}Z_{\beta_2} \\ &= \chi\left(-2^{-1}(\alpha_1\beta_1+\alpha_2\beta_2)\right) X_{\alpha_1}\left(\chi(\alpha_2\beta_1)X_{\alpha_2}Z_{\beta_1}\right)Z_{\beta_2} \\ &= \chi\left(-2^{-1}(\alpha_1\beta_1+\alpha_2\beta_2)+\alpha_2\beta_1\right) X_{\alpha_1+\alpha_2}Z_{\beta_1+\beta_2}. \end{aligned} \tag{45}$$

Notice the algebraic identity:

$$-2^{-1}(\alpha_1\beta_1+\alpha_2\beta_2)+\alpha_2\beta_1 = -2^{-1}(\alpha_1+\alpha_2)(\beta_1+\beta_2) + 2^{-1}(\alpha_1\beta_2-\alpha_2\beta_1). \tag{46}$$

Factoring out the phase term:

$$\begin{aligned} D(\alpha_1,\beta_1)D(\alpha_2,\beta_2) &= \chi\left(2^{-1}(\alpha_1\beta_2-\alpha_2\beta_1)\right)\left[\chi\left(-2^{-1}(\alpha_1+\alpha_2)(\beta_1+\beta_2)\right) X_{\alpha_1+\alpha_2}Z_{\beta_1+\beta_2}\right] \\ &= \chi\left(2^{-1}(\alpha_1\beta_2-\alpha_2\beta_1)\right) D(\alpha_1+\alpha_2,\beta_1+\beta_2). \end{aligned} \tag{47}$$

□

Theorem 4.4 (Orthogonality of the Displacement Operator Basis)**.** *The set of q^2 displacement operators $\{D(\alpha,\beta) : \alpha,\beta \in \mathbb{F}_q\}$ forms an orthogonal basis for the operator algebra $\mathcal{B}(\mathcal{H}_q)$, satisfying:*

$$\mathrm{Tr}\left(D(\alpha,\beta)^\dagger D(\alpha',\beta')\right) = q\,\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}. \tag{48}$$

Proof. Compute $D(\alpha,\beta)^\dagger$:

$$\begin{aligned} D(\alpha,\beta)^\dagger &= \chi\left(2^{-1}\alpha\beta\right) Z_\beta^\dagger X_\alpha^\dagger = \chi\left(2^{-1}\alpha\beta\right) Z_{-\beta}X_{-\alpha} \\ &= \chi\left(2^{-1}\alpha\beta\right)\chi(\alpha\beta)X_{-\alpha}Z_{-\beta} = \chi\left(-2^{-1}(-\alpha)(-\beta)\right) X_{-\alpha}Z_{-\beta} = D(-\alpha,-\beta). \end{aligned} \tag{49}$$

By Theorem 4.3:

$$\begin{aligned} D(\alpha,\beta)^\dagger D(\alpha',\beta') &= D(-\alpha,-\beta)D(\alpha',\beta') \\ &= \chi\left(2^{-1}(-\alpha\beta'-(-\beta)\alpha')\right) D(\alpha'-\alpha,\beta'-\beta) \\ &= \chi\left(2^{-1}(\alpha\beta'-\alpha'\beta)\right)\chi\left(-2^{-1}(\alpha'-\alpha)(\beta'-\beta)\right) X_{\alpha'-\alpha}Z_{\beta'-\beta}. \end{aligned} \tag{50}$$

Taking the trace:

$$\mathrm{Tr}\left(X_{\alpha'-\alpha}Z_{\beta'-\beta}\right) = \sum_{\xi\in\mathbb{F}_q}\langle\xi|X_{\alpha'-\alpha}Z_{\beta'-\beta}|\xi\rangle = \sum_{\xi\in\mathbb{F}_q}\chi((\beta'-\beta)\xi)\underbrace{\langle\xi|\xi+\alpha'-\alpha\rangle}_{\delta_{\alpha',\alpha}}. \tag{51}$$

If $\alpha' \neq \alpha$, every term in the sum vanishes due to the delta function, yielding zero. If $\alpha' = \alpha$, the expression reduces to $\sum_{\xi\in\mathbb{F}_q}\chi((\beta'-\beta)\xi)$, which by Theorem 2.5 evaluates to $q\,\delta_{\beta',\beta}$. Combining both conditions yields $q\,\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}$. □

5 Discrete Phase Space Affine Geometry and Striations

The finite phase space is formally modeled as the affine plane $\Gamma = \mathbb{F}_q \times \mathbb{F}_q \cong \mathbb{F}_q^2$.

Definition 5.1 (Affine Lines and Striations)**.** An affine line $L(a, b, c)$ in Γ is the solution set of the linear equation:

$$L(a,b,c) \equiv \left\{(\alpha,\beta) \in \mathbb{F}_q^2 : a\alpha + b\beta = c\right\}, \quad (a,b) \in \mathbb{F}_q^2 \setminus \{(0,0)\},\ c \in \mathbb{F}_q. \tag{52}$$

Two lines $L(a_1, b_1, c_1)$ and $L(a_2, b_2, c_2)$ are parallel ($L_1 \parallel L_2$) if their normal vectors (a_1, b_1) and (a_2, b_2) are linearly dependent over $\mathbb{F}_q$, meaning $a_1 b_2 - a_2 b_1 = 0$.

A *striation* $\mathcal{S}$ is an equivalence class of parallel lines:

$$\mathcal{S}_{(a:b)} = \{L(a,b,c) : c \in \mathbb{F}_q\}, \quad (a:b) \in \mathbb{P}^1(\mathbb{F}_q). \tag{53}$$

Theorem 5.1 (Enumeration and Incidence Axioms of $\mathbb{F}_q^2$)**.** *The affine plane $\Gamma = \mathbb{F}_q^2$ satisfies:*

1. *Every line L contains exactly q points.*
2. *There are exactly $q + 1$ distinct striations, parameterized by the projective line $\mathbb{P}^1(\mathbb{F}_q)$:*
   - *q striations with finite slope $s \in \mathbb{F}_q$: lines $\beta = s\alpha + c$ ($c \in \mathbb{F}_q$);*
   - *1 vertical striation $s = \infty$: lines $\alpha = c$ ($c \in \mathbb{F}_q$).*
3. *Each striation partitions the q^2 points of Γ into q mutually disjoint parallel lines.*
4. *Any two non-parallel lines $L_1 \nparallel L_2$ intersect in exactly one point: $|L_1 \cap L_2| = 1$.*

Proof. 1. Let $a\alpha + b\beta = c$ with $(a, b) \neq (0, 0)$. If $b \neq 0$, then for each of the q choices of $\alpha \in \mathbb{F}_q$, $\beta = b^{-1}(c - a\alpha)$ is uniquely determined. Hence $|L| = q$.

2. Two non-zero vectors (a, b) and (a', b') define the same parallel class if and only if $(a', b') = \lambda(a, b)$ for some $\lambda \in \mathbb{F}_q^\times$. The number of lines through the origin is therefore $(q^2 - 1)/(q - 1) = q + 1$.

3. For a fixed normal $(a, b) \neq (0, 0)$, the mapping $(\alpha, \beta) \mapsto a\alpha + b\beta$ is an epimorphism from $\mathbb{F}_q^2$ to $\mathbb{F}_q$. Its fibers are the lines $L(a, b, c)$, forming a partition of $\mathbb{F}_q^2$.

4. Consider the system $a_1\alpha + b_1\beta = c_1$ and $a_2\alpha + b_2\beta = c_2$. Its coefficient matrix has determinant $\Delta = a_1 b_2 - a_2 b_1$. Since $L_1 \nparallel L_2$, $\Delta \neq 0$. By Cramer's rule, there is a unique solution $(\alpha, \beta) \in \mathbb{F}_q^2$.

□

6 Mutually Unbiased Bases (MUBs) via Affine Striations

Definition 6.1 (Mutually Unbiased Bases)**.** Two orthonormal bases $\mathcal{B}_1 = \{|\phi_j\rangle\}_{j=1}^q$ and $\mathcal{B}_2 = \{|\psi_k\rangle\}_{k=1}^q$ of $\mathcal{H}_q$ are mutually unbiased if:

$$|\langle\phi_j|\psi_k\rangle|^2 = \frac{1}{q}, \quad \forall j, k \in \{1, \ldots, q\}. \tag{54}$$

A set of bases $\{\mathcal{B}_0, \mathcal{B}_1, \ldots, \mathcal{B}_m\}$ is a mutually unbiased set if every pair is mutually unbiased.

Theorem 6.1 (MUB Construction from Striations in Dimension $q = p^n$)**.** *Let $p > 2$. For each slope $s \in \mathbb{F}_q$, define the operator $V_s \in \mathcal{U}(\mathcal{H}_q)$ by:*

$$V_s \equiv \sum_{\xi \in \mathbb{F}_q} \chi\left(2^{-1} s \xi^2\right) |\xi\rangle\langle\xi|. \tag{55}$$

Then the $q+1$ bases defined by:

$$\mathcal{B}_\infty \equiv \{|\alpha\rangle : \alpha \in \mathbb{F}_q\} \quad \textit{(Computational / Position Basis)}, \tag{56}$$

$$\mathcal{B}_s \equiv \{|\psi_{s,\beta}\rangle : \beta \in \mathbb{F}_q\}, \quad |\psi_{s,\beta}\rangle \equiv V_s \mathcal{F}_q |\beta\rangle = \frac{1}{\sqrt{q}} \sum_{\xi \in \mathbb{F}_q} \chi\left(2^{-1} s \xi^2 + \beta \xi\right) |\xi\rangle, \tag{57}$$

form a complete, maximal set of $q+1$ Mutually Unbiased Bases in $\mathcal{H}_q$.

Proof. First, check orthonormality within a given basis $\mathcal{B}_s$. Since V_s and $\mathcal{F}_q$ are unitary, their product $V_s \mathcal{F}_q$ is unitary. Thus $\mathcal{B}_s = \{V_s \mathcal{F}_q |\beta\rangle\}$ is an orthonormal basis:

$$\langle \psi_{s,\beta} | \psi_{s,\beta'} \rangle = \langle \beta | \mathcal{F}_q^\dagger V_s^\dagger V_s \mathcal{F}_q | \beta' \rangle = \langle \beta | \beta' \rangle = \delta_{\beta,\beta'}. \tag{58}$$

Second, check mutual unbiasedness between $\mathcal{B}_\infty$ and $\mathcal{B}_s$:

$$|\langle \alpha | \psi_{s,\beta} \rangle|^2 = \left| \frac{1}{\sqrt{q}} \chi\left(2^{-1} s \alpha^2 + \beta \alpha\right) \right|^2 = \frac{1}{q} \left| \omega_p^{\mathrm{Tr}\left(2^{-1} s \alpha^2 + \beta \alpha\right)} \right|^2 = \frac{1}{q}. \tag{59}$$

Third, check mutual unbiasedness between two distinct bases $\mathcal{B}_s$ and $\mathcal{B}_{s'}$ with $s \neq s'$:

$$\begin{aligned} \langle \psi_{s,\beta} | \psi_{s',\beta'} \rangle &= \left(\frac{1}{\sqrt{q}} \sum_{\xi \in \mathbb{F}_q} \chi\left(-2^{-1} s \xi^2 - \beta \xi\right) \langle \xi | \right) \left(\frac{1}{\sqrt{q}} \sum_{\eta \in \mathbb{F}_q} \chi\left(2^{-1} s' \eta^2 + \beta' \eta\right) |\eta\rangle \right) \\ &= \frac{1}{q} \sum_{\xi \in \mathbb{F}_q} \chi\left(2^{-1}(s' - s)\xi^2 + (\beta' - \beta)\xi\right). \end{aligned} \tag{60}$$

Let $\Delta s = s' - s \neq 0$ and $\Delta\beta = \beta' - \beta$. Completing the square over $\mathbb{F}_q$ (recalling $p \neq 2$):

$$\begin{aligned} 2^{-1} \Delta s\, \xi^2 + \Delta\beta\, \xi &= 2^{-1} \Delta s \left(\xi^2 + 2(\Delta s)^{-1} \Delta\beta\, \xi\right) \\ &= 2^{-1} \Delta s \left(\xi + (\Delta s)^{-1} \Delta\beta\right)^2 - 2^{-1} (\Delta s)^{-1} (\Delta\beta)^2. \end{aligned} \tag{61}$$

Factoring out the constant phase:

$$\langle \psi_{s,\beta} | \psi_{s',\beta'} \rangle = \frac{1}{q} \chi\left(-2^{-1}(\Delta s)^{-1}(\Delta\beta)^2\right) \sum_{\xi \in \mathbb{F}_q} \chi\left(2^{-1} \Delta s \left(\xi + (\Delta s)^{-1} \Delta\beta\right)^2\right). \tag{62}$$

Under the variable shift $\xi' = \xi + (\Delta s)^{-1} \Delta\beta$, the summation becomes the quadratic Gauss sum over the finite field $\mathbb{F}_q$:

$$G(\Delta s) = \sum_{\xi' \in \mathbb{F}_q} \chi\left(2^{-1} \Delta s\, (\xi')^2\right). \tag{63}$$

By classical finite field Gauss sum theory, since $2^{-1}\Delta s \neq 0$, the modulus of the quadratic character sum evaluates to:

$$|G(\Delta s)| = \sqrt{q}. \tag{64}$$

Substituting this back into the transition amplitude:

$$|\langle \psi_{s,\beta} | \psi_{s',\beta'} \rangle| = \frac{1}{q} \cdot 1 \cdot \sqrt{q} = \frac{1}{\sqrt{q}} \implies |\langle \psi_{s,\beta} | \psi_{s',\beta'} \rangle|^2 = \frac{1}{q}. \tag{65}$$

Thus, all $q+1$ bases are strictly mutually unbiased. □

7 Discrete Wigner Function and Phase Space Point Operators

We now construct the discrete phase space representation of quantum density matrices.

Definition 7.1 (Phase Space Point Operators)**.** For odd p, the Stratonovich-Weyl point operator $A(\alpha, \beta)$ associated with coordinate $(\alpha, \beta) \in \mathbb{F}_q^2$ is defined by:

$$A(\alpha, \beta) \equiv \frac{1}{q} \sum_{u,v \in \mathbb{F}_q} \chi(u\beta - v\alpha) D(u, v). \tag{66}$$

Equivalently, under the Wootters-Gibbons-Hoffman geometric formulation:

$$A(\alpha, \beta) \equiv \sum_{L \ni (\alpha,\beta)} P(L) - I_q, \tag{67}$$

where the sum runs over the $q + 1$ lines passing through (α, β), and $P(L)$ is the projection operator onto the quantum state associated with line L.

Theorem 7.1 (Stratonovich-Weyl Axioms of Point Operators)**.** *The point operators $A(\alpha, \beta)$ satisfy:*

1. *Hermiticity:* $A(\alpha, \beta)^\dagger = A(\alpha, \beta)$*;*
2. *Unit Trace:* $\mathrm{Tr}(A(\alpha, \beta)) = 1$*;*
3. *Tracial Orthogonality:* $\mathrm{Tr}(A(\alpha, \beta)A(\alpha', \beta')) = q\, \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}$.

Proof. 1. Since $D(u, v)^\dagger = D(-u, -v)$:

$$\begin{aligned} A(\alpha, \beta)^\dagger &= \frac{1}{q} \sum_{u,v \in \mathbb{F}_q} \chi(-(u\beta - v\alpha)) D(-u, -v) \\ &= \frac{1}{q} \sum_{u',v' \in \mathbb{F}_q} \chi(u'\beta - v'\alpha) D(u', v') = A(\alpha, \beta). \end{aligned} \tag{68}$$

2. Taking the trace:

$$\mathrm{Tr}(A(\alpha, \beta)) = \frac{1}{q} \sum_{u,v \in \mathbb{F}_q} \chi(u\beta - v\alpha) \mathrm{Tr}(D(u, v)). \tag{69}$$

From Theorem 4.4, $\mathrm{Tr}(D(u, v)) = q\, \delta_{u,0} \delta_{v,0}$. Therefore:

$$\mathrm{Tr}(A(\alpha, \beta)) = \frac{1}{q} \chi(0) \cdot q = 1. \tag{70}$$

3. Using the operator orthogonality of Theorem 4.4:

$$\begin{aligned} \mathrm{Tr}\left(A(\alpha, \beta)A(\alpha', \beta')\right) &= \frac{1}{q^2} \sum_{u,v,u',v'} \chi(u\beta - v\alpha)\chi(u'\beta' - v'\alpha') \mathrm{Tr}(D(u, v)D(u', v')) \\ &= \frac{1}{q^2} \sum_{u,v} \chi(u\beta - v\alpha)\chi(-u\beta' + v\alpha') q \\ &= \frac{1}{q} \left(\sum_{u \in \mathbb{F}_q} \chi(u(\beta - \beta')) \right) \left(\sum_{v \in \mathbb{F}_q} \chi(v(\alpha' - \alpha)) \right) \\ &= \frac{1}{q} (q\, \delta_{\beta,\beta'})(q\, \delta_{\alpha,\alpha'}) = q\, \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}. \end{aligned} \tag{71}$$

□

Definition 7.2 (Discrete Wigner Quasi-Probability Distribution)**.** For any density operator $\rho \in \mathcal{S}(\mathcal{H}_q)$, the discrete Wigner function $W_\rho : \mathbb{F}_q^2 \to \mathbb{R}$ is defined by:

$$W_\rho(\alpha, \beta) \equiv \frac{1}{q} \mathrm{Tr}\left(\rho A(\alpha, \beta)\right). \tag{72}$$

Theorem 7.2 (State Reconstruction and Marginality)**.** *The discrete Wigner function satisfies:*

1. *Inversion / State Reconstruction:*

$$\rho = \sum_{\alpha,\beta \in \mathbb{F}_q} W_\rho(\alpha, \beta) A(\alpha, \beta); \tag{73}$$

2. *Total Normalization:*

$$\sum_{\alpha,\beta \in \mathbb{F}_q} W_\rho(\alpha, \beta) = 1; \tag{74}$$

3. *Line Marginality: For any affine line $L \subset \mathbb{F}_q^2$:*

$$\sum_{(\alpha,\beta) \in L} W_\rho(\alpha, \beta) = \mathrm{Tr}(\rho P(L)), \tag{75}$$

where $P(L)$ is the projection operator corresponding to the quantum state identified with line L.

Proof. 1. Since $\{A(\alpha, \beta)\}$ forms an orthogonal operator basis with $\mathrm{Tr}(A(\alpha, \beta)A(\alpha', \beta')) = q\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}$, expanding $\rho = \sum_{\alpha,\beta} c_{\alpha,\beta} A(\alpha, \beta)$ and taking the trace against $A(\alpha', \beta')$ yields $c_{\alpha',\beta'} = \frac{1}{q}\mathrm{Tr}(\rho A(\alpha', \beta')) = W_\rho(\alpha', \beta')$.

2. Summing over all points:

$$\sum_{\alpha,\beta} W_\rho(\alpha, \beta) = \frac{1}{q} \mathrm{Tr}\left(\rho \sum_{\alpha,\beta} A(\alpha, \beta)\right) = \frac{1}{q} \mathrm{Tr}\left(\rho \cdot q I_q\right) = \mathrm{Tr}(\rho) = 1. \tag{76}$$

3. Using the geometric identity $\sum_{(\alpha,\beta) \in L} A(\alpha, \beta) = qP(L)$, which arises from the inverse Radon transform on $\mathbb{F}_q^2$:

$$\sum_{(\alpha,\beta) \in L} W_\rho(\alpha, \beta) = \frac{1}{q} \mathrm{Tr}\left(\rho \sum_{(\alpha,\beta) \in L} A(\alpha, \beta)\right) = \frac{1}{q} \mathrm{Tr}(\rho \cdot qP(L)) = \mathrm{Tr}(\rho P(L)). \tag{77}$$

□

Theorem 7.3 (Discrete Hudson's Theorem and Negativity as Quantum Magic)**.** *A pure quantum state $|\psi\rangle \in \mathcal{H}_q$ has a strictly non-negative discrete Wigner function $W_{|\psi\rangle}(\alpha, \beta) \geq 0$ for all $(\alpha, \beta) \in \mathbb{F}_q^2$ if and only if $|\psi\rangle$ is a stabilizer state (an eigenstate of an abelian subgroup of the Clifford group). Any non-stabilizer state exhibits regions of negativity:*

$$\mathcal{M}(|\psi\rangle) \equiv \sum_{\alpha,\beta \in \mathbb{F}_q} \max\left(0, -W_{|\psi\rangle}(\alpha, \beta)\right) > 0, \tag{78}$$

which strictly quantifies the non-Clifford resource of quantum magic required for universal fault-tolerant quantum computation.

8 The $\mathbb{F}_4$ Symplectic Isomorphism for Multi-Qubit Systems

When modeling registers of n qubits, the system dimension is $d = 2^n$. The base field is $\mathbb{F}_2 = \{0, 1\}$. While multi-qubit Pauli operators carry phase factors $\{\pm 1, \pm i\}$, their projective quotient possesses a canonical field structure over $\mathbb{F}_4$.

Definition 8.1 (The Field $\mathbb{F}_4$)**.** The finite field $\mathbb{F}_4 = \mathrm{GF}(4)$ consists of four elements:

$$\mathbb{F}_4 = \{0, 1, \omega, \bar{\omega} = \omega^2\}, \tag{79}$$

governed by the irreducible polynomial $P(x) = x^2 + x + 1 \in \mathbb{F}_2[x]$, imposing:

$$\omega^2 + \omega + 1 = 0 \implies \omega^2 = \omega + 1, \quad \omega^3 = 1. \tag{80}$$

Definition 8.2 (Pauli-to-$\mathbb{F}_4$ Isomorphism)**.** Let $\mathcal{P}_1 = \{\pm I, \pm iI, \pm X, \pm iX, \pm Y, \pm iY, \pm Z, \pm iZ\}$ be the single-qubit Pauli group. The quotient group by its center $\mathcal{Z}(\mathcal{P}_1) = \{\pm I, \pm iI\}$ is isomorphic to the additive group of $\mathbb{F}_4$:

$$\phi : \mathcal{P}_1/\mathcal{Z}(\mathcal{P}_1) \xrightarrow{\sim} \mathbb{F}_4, \tag{81}$$

defined by the mapping:

$$I \mapsto 0, \quad X \mapsto 1, \quad Z \mapsto \omega, \quad Y \mapsto \omega + 1 = \omega^2. \tag{82}$$

Theorem 8.1 (Multi-Qubit Pauli Commutation via Symplectic Form)**.** *Let $P_u, P_v \in \mathcal{P}_n$ be n-qubit Pauli operators mapped to vectors $u, v \in \mathbb{F}_4^n$. The absolute trace map on $\mathbb{F}_4/\mathbb{F}_2$ is $\mathrm{Tr}_4(\xi) = \xi + \xi^2$. Define the trace Hermitian inner product on $\mathbb{F}_4^n$:*

$$(u, v)_{\mathrm{symp}} \equiv \mathrm{Tr}_4\left(\sum_{j=1}^{n} u_j \bar{v}_j\right) \in \mathbb{F}_2. \tag{83}$$

Then P_u and P_v commute if and only if their symplectic product vanishes:

$$[P_u, P_v] = 0 \iff (u, v)_{\mathrm{symp}} = 0. \tag{84}$$

Proof. Decompose each component $u_j = x_j + \omega z_j$ where $x_j, z_j \in \mathbb{F}_2$. Then u_j represents $X^{x_j} Z^{z_j}$. Compute $u_j \bar{v}_j$:

$$\begin{aligned} u_j \bar{v}_j &= (x_j + \omega z_j)(x'_j + \omega^2 z'_j) = x_j x'_j + \omega^2 x_j z'_j + \omega z_j x'_j + \omega^3 z_j z'_j \\ &= (x_j x'_j + z_j z'_j) + \omega z_j x'_j + \omega^2 x_j z'_j. \end{aligned} \tag{85}$$

Applying the trace $\mathrm{Tr}_4(\xi) = \xi + \xi^2$:

$$\begin{aligned} \mathrm{Tr}_4(u_j \bar{v}_j) &= \mathrm{Tr}_4\left(x_j x'_j + z_j z'_j\right) + \mathrm{Tr}_4(\omega) z_j x'_j + \mathrm{Tr}_4(\omega^2) x_j z'_j \\ &= 0 + (\omega + \omega^2) z_j x'_j + (\omega^2 + \omega) x_j z'_j \\ &= 1 \cdot (z_j x'_j + x_j z'_j) = x_j z'_j + z_j x'_j \pmod 2. \end{aligned} \tag{86}$$

Summing over all n qubits:

$$(u, v)_{\mathrm{symp}} = \sum_{j=1}^{n} (x_j z'_j + z_j x'_j) \pmod 2. \tag{87}$$

In standard quantum mechanics, $X_{x_j} Z_{z_j} X_{x'_j} Z_{z'_j} = (-1)^{x_j z'_j + z_j x'_j} X_{x'_j} Z_{z'_j} X_{x_j} Z_{z_j}$. Multiplying across all n tensor factors gives:

$$P_u P_v = (-1)^{(u,v)_{\mathrm{symp}}} P_v P_u. \tag{88}$$

Thus, $[P_u, P_v] = 0$ if and only if $(u, v)_{\mathrm{symp}} \equiv 0 \pmod 2$. □

9 Classical Coding and Hardware-Accelerated Transforms

The GFFT is identical to the discrete Fourier transform over finite fields utilized in Reed-Solomon (RS) error-correcting codes.

Theorem 9.1 (Reed-Solomon Codeword Generation as GFFT)**.** *Let $m(x) = \sum_{j=0}^{k-1} m_j x^j$ be a message polynomial of length k over $\mathbb{F}_q$. Let γ be a primitive element of $\mathbb{F}_q^\times$. Evaluating $m(x)$ at all powers γ^ℓ ($\ell = 0, \ldots, q-2$) produces the codeword:*

$$c_\ell = m(\gamma^\ell) = \sum_{j=0}^{k-1} m_j (\gamma^\ell)^j = \sum_{j=0}^{k-1} m_j \gamma^{j\ell}. \tag{89}$$

This is precisely the forward Galois Field Fourier Transform of the zero-padded message vector $m \in \mathbb{F}_q^{q-1}$.

Theorem 9.2 (BCH Bound and Spectral Consecutive Zero Run)**.** *If the GFFT spectrum $C = \mathcal{F}_q(c)$ of a codeword c contains a run of at least $d-1$ consecutive zeros:*

$$C_{\ell_0} = C_{\ell_0+1} = \cdots = C_{\ell_0+d-2} = 0, \tag{90}$$

then the code has minimum Hamming distance at least d, and is capable of correcting up to $t = \lfloor (d-1)/2 \rfloor$ arbitrary errors.

Remark 9.1 (Hardware Acceleration via AVX-512 `GF2P8AFFINEINVQB`). Modern x86-64 microprocessors execute parallel Galois field arithmetic via the AVX-512 instructions `GF2P8AFFINEINVQB` and `GF2P8MULB`. The instruction performs affine transformations over $\mathbb{F}_{2^8}$:

$$y_i = A \cdot x_i^{-1} \oplus b, \tag{91}$$

where $A \in \mathbb{F}_2^{8\times 8}$ is an 8-bit binary matrix, $x_i \in \mathbb{F}_{2^8}$, and $b \in \mathbb{F}_2^8$. This allows vector registers (512-bit width) to process 64 Galois field operations per clock cycle, accelerating cryptographic inversion (AES S-box) and parallel GFFT decoding.

10 Non-Archimedean p-Adic Transforms and Galois Field Quantum Theory

10.1 p-Adic Quantum Kinematics

Beyond finite fields $\mathbb{F}_q$, the non-Archimedean completion of rational numbers $\mathbb{Q}$ with respect to the p-adic absolute value $|x|_p$ yields the p-adic field $\mathbb{Q}_p$.

Definition 10.1 (p-Adic Integral Transform)**.** For $f \in L^2(\mathbb{Q}_p)$, the p-adic Fourier transform is defined using the fractional part character:

$$\mathcal{F}_{\mathbb{Q}_p}[f](\xi) = \int_{\mathbb{Q}_p} \chi_p(\xi x) f(x)\, dx = \int_{\mathbb{Q}_p} \exp(2\pi i \{\xi x\}_p) f(x)\, dx. \tag{92}$$

Theorem 10.1 (Power-Law Propagator Decay vs Exponential Tunneling)**.** *Unlike continuum quantum mechanics over $\mathbb{R}$ where propagators in classically forbidden regimes decay exponentially $K(x,t) \sim \exp(-\kappa|x|)$, the p-adic quantum propagator satisfies an ultrametric power-law decay:*

$$|K(x,t)|_p \sim |x|_p^{-\alpha}, \quad \text{as } |x|_p \to \infty, \tag{93}$$

demonstrating that non-Archimedean phase spaces sustain long-range quantum correlations without continuous limits.

10.2 Galois Field Quantum Theory (GFQT) Cosmologies

In Felix Lev's Galois Field Quantum Theory (GFQT), the continuum is entirely abandoned at the fundamental level:

1. Physical reality is defined strictly over a finite field $\mathbb{F}_p$ of prime characteristic $p \sim 10^{120}$, where p is directly linked to the de Sitter entropy of the cosmological horizon:
$$p \sim \exp\left(\frac{3\pi c^3}{G\hbar\Lambda}\right) \sim 10^{122}. \tag{94}$$
2. The Poincaré group is replaced by the modular de Sitter group $\mathrm{SO}(1,4)$ formulated over $\mathbb{F}_p$.
3. The hierarchy problem is naturally resolved: the gravitational coupling constant is kinematically determined by the characteristic $G \sim 1/p$, naturally explaining why gravitational interactions are 10^{40} times weaker than electrodynamics without arbitrary parameter fine-tuning.

11 Summary of Mathematical Theorems and Structural Correspondence

Table 1 synthesizes the mathematical correspondence between continuum quantum mechanics and the finite Galois Field Mechanic Transform.

Table 1: **Comprehensive Structural Correspondence Between Continuum and Galois Field Mechanics.**

Kinematical Property	**Continuum Quantum Mechanics ($\mathbb{R}^{2n}$)**	**Galois Field Transform ($\mathbb{F}_q^2$, $q = p^n$)**
Phase Space Substrate	Continuous manifold $\mathbb{R} \times \mathbb{R}$	Discrete affine torus $\mathbb{F}_q \times \mathbb{F}_q$
Additive Character	$\chi(x) = e^{ix}$	$\chi(\alpha) = \exp\left(\frac{2\pi i}{p}\mathrm{Tr}(\alpha)\right)$
Fourier Transform Operator	$\mathcal{F} = \frac{1}{\sqrt{2\pi}} \int e^{ixp} \lvert p\rangle\langle x\rvert dp$	$\mathcal{F}_q = \frac{1}{\sqrt{q}} \sum_{\alpha,\beta} \chi(\alpha\beta) \lvert\beta\rangle\langle\alpha\rvert$
Kinematic Commutation	$[x, p] = i\hbar I$	$Z_\beta X_\alpha = \chi(\alpha\beta) X_\alpha Z_\beta$
MUB Saturation	Unbounded continuous bases	Exactly $q+1$ MUBs via striations
Point Operator $A(x,p)$	Parity displaced: $D(x,p)\Pi D(x,p)^\dagger$	$\frac{1}{q}\sum_{u,v} \chi(up - vx) D(u,v)$
Wigner Marginality	$\int W(x,p)dp = \lvert\psi(x)\rvert^2$	$\sum_{(\alpha,\beta)\in L} W(\alpha,\beta) = \mathrm{Tr}(\rho P(L))$
Hudson's Positivity	Pure Gaussian wavepackets only	Clifford stabilizer states only
Computational Resource	Squeezing / Non-Gaussianity	Quantum Magic (Wigner negativity)
Multi-Qubit Algebraic Ring	Tensor product $\mathcal{H}_2^{\otimes n}$	$\mathbb{F}_4^n$ Symplectic Linear Algebra